RIKRITI KOIRALA
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Programming Languages	JAVA, J2EE, C, C++, Python.
Frameworks	Spring, Hibernate, Spring MVC, Spring IOC, Spring AOP, Spring Security, Spring Data, Struts, Spring Batch, Spring Boot.
Java Technologies	Core Java, J2EE, Servlets, JSP, JDBC, Collections, jQuery.
Web/XML Technologies	HTML5, CSS3, JavaScript, jQuery, Bootstrap, RESTFUL, Angular JS, React, AJAX, NodeJS, JSON, XML, DOM, XSD and XSLT.
Cloud Technologies	Azure DevOps, AWS (EC2, S3, IAM, RDS)
Databases	Oracle9i/10G/11g, SQL-Server, MySQL server, MS SQL, IBM DB2, MongoDB, NoSQL.
Web/Application Servers	Apache Tomcat, IBM WebSphere, WebLogic, Glassfish, Web logic Application server, JBOSS.
Build Automation Tools	ANT, Maven, Gradle, Docker, Log4J, and Jenkins
IDE / Tools	Eclipse, IntelliJ, Spring Tool Suite (STS) and NetBeans
Testing Tools/ Others	JUnit, Mockito, Soap UI, Putty, Cucumber and JIRA
Version Control	Tortoise SVN, and GIT
Platforms	Windows, Mac, Linux, and Unix.
Methodologies	Agile, Waterfall, Kanban, Test Driven Development

PROFESSIONAL EXPERIENCE:

Bank of America, Charlotte, NC
June 2025 – Present | Backend Java Developer

- Designed and developed server-side logic, APIs, and data integration layers to facilitate seamless communication between applications and services.
- Experienced with Java Collections Framework, efficiently utilizing data structures such as lists, sets, maps, and queues to optimize storage and retrieval operations.
- Worked with infrastructure-as-code (IaC) tools, including AWS CloudFormation and Terraform, to define, provision, and manage cloud resources programmatically, ensuring efficient deployment and configuration.
- Worked with Python libraries such as NumPy, Pandas, and Requests for data manipulation and API interactions. Integrated Python with databases like MongoDB, PostgreSQL, and MySQL.
- Specialized in server-side development, focusing on designing and implementing RESTful APIs and leveraging asynchronous programming techniques to enhance performance and scalability.
- Hands-on expertise with AWS Lambda, enabling event-driven architectures by executing functions without managing underlying server infrastructure.
- Built and deployed serverless functions using AWS Lambda to handle events triggered by HTTP requests, S3 operations, and DynamoDB changes, facilitating scalable microservices-based applications.

- Strong command of version control systems, particularly Git, along with experience in setting up and maintaining CI/CD pipelines to automate software delivery, conduct code reviews, and enforce quality standards.
- Leveraged Spring Boot to streamline development through auto-configuration, reducing manual setup and minimizing boilerplate code, resulting in accelerated application prototyping and iteration.
- Implemented microservices architecture with Spring Boot to develop modular, independently deployable services, improving scalability and manageability of complex systems.
- Integrated Apache Kafka as a distributed messaging system, establishing a publish-subscribe model where producers send messages to topics and consumers process them in real time.
- Designed and implemented NoSQL database solutions using Amazon DynamoDB, creating tables, defining schemas, and performing CRUD operations for optimized data storage and retrieval.
- Scaled Kafka clusters horizontally by dynamically adding or removing brokers to accommodate fluctuating workloads, leveraging partition rebalancing to maintain efficient data distribution.
- Applied Scala's functional programming features—including immutability, higher-order functions, pattern matching, and type inference—to write concise, expressive, and maintainable code. Proficient in integrating Scala and Java, enabling seamless interoperability between Scala modules and existing Java frameworks and libraries.
- Utilized Dagger for dependency injection, streamlining dependency management in Java and Android applications while reducing boilerplate code.
- Developed applications using Angular's component-based architecture to enhance modularity, maintainability, and reusability of code. Implemented state management in Angular applications using NgRx, adopting a Redux-inspired approach to efficiently handle data flow and application state.
- Leveraged Dagger's constructor injection to automatically manage dependencies, improving code readability and maintainability.
- Developed asynchronous, event-driven backend services using Node.js, leveraging the non-blocking I/O model for high-performance applications.
- Familiar with Apache Spark's real-time processing capabilities, particularly Spark Streaming, for ingesting and analyzing live data streams in real-time applications.
- Utilized Kafka Connect to integrate Kafka with external systems such as PostgreSQL, MongoDB, Elasticsearch, and Redis.
- Integrated JWT / OAuth2 authentication in Swagger by configuring security schemes in OpenAPI definitions. Enhanced API validation by incorporating constraints, response schemas, and API contract validation via Swagger Request Validator.
- Designed and managed replica sets for high availability and sharding for horizontal scaling using MongoDB.
- Experienced in containerization, creating Docker images by packaging applications with dependencies and required configurations for streamlined deployment.
- Designed reusable infrastructure components using Terraform modules, promoting code modularity and maintainability across multiple projects.
- Designed and implemented ETL pipelines to efficiently load, transform, and process data into Redshift using AWS Glue, Lambda, and Python scripts.
- Managed Terraform state files to track infrastructure changes over time, utilizing remote state backends such as Amazon S3 and Terraform Cloud/Enterprise for centralized state management and team collaboration.
- Utilized Kubernetes for orchestrating containerized applications, automating deployment, scaling, and operational management of microservices across distributed clusters.

Environment: Java, Server-Side Logic, Restful API, Terraform, Asynchronous Programming, Server Architecture, Node.js, Apache Kafka, DynamoDB, Cluster Scaling, Swagger, Functional programming, Dagger, Angular, Docker, S3, EC2, Kubernetes

Northwest Missouri State University, Maryville - MO (Part-time)
Oct 2023 - Dec 2024 | Software Developer – Java

- Engaged in requirements gathering, analysis, design, business logic implementation, unit testing, deployment, and application maintenance.
- Followed Agile methodology throughout the development process.
- Implemented design patterns such as Singleton, Business Delegate, Value Object, Session Façade, Service Locator, DAO, DTO, and MVC.
- Utilized Angular 10 controllers to manage page data and modules for binding data between the user interface and controller. Made REST API calls using Angular 10's HTTP and resource services.
- Built reusable components, custom modules, directives, pipes, and services in Angular. Also created custom Angular directives to develop reusable components for use across multiple application pages.
- Configured Docker containers and wrote Docker files for different environments.
- Deployed Spring Boot-based microservices in Docker and Amazon EC2 containers via the AWS management console.
- Modernized an existing application by re-architecting it with Java 11, Spring Boot, the Spring Reactive Stack (WebFlux), PostgreSQL, and Maven.
- Designed and developed Kafka-based event-driven architectures to handle high-throughput data processing.
- Integrated JMS for asynchronous data exchange between J2EE components and legacy systems.
- Built server-side functionality to interact with the database using Spring Boot and Hibernate.
- Worked with Vagrant, AWS, and Kubernetes-based container deployments to create self-contained environments for development teams and streamline containerization for releases.
- Developed DevOps pipelines using OpenShift and Kubernetes within a microservices architecture.
- Automated data processing and ETL pipelines using Python and Pandas.
- Integrated MongoDB with Java applications using Spring Data MongoDB.
- Migrated a monolithic application to a microservices architecture using Spring Boot, adhering to the 12-factor app methodology. Deployed, scaled, and configured microservices in PCF, writing manifest files as needed.
- Integrated Swagger with API Gateway (AWS API Gateway) to expose and manage API endpoints with standardized documentation. Defined API contracts using YAML/JSON-based OpenAPI Specification (OAS) for standardized request/response models.
- Utilized AWS CloudWatch and Redshift system tables for monitoring cluster performance and troubleshooting issues.
- Configured and optimized OpenShift routes, services, and deployments using YAML manifests and Helm charts for efficient application lifecycle management.
- Wrote automated unit test cases for microservices using JUnit, Mockito, and Sonar, integrating them into the Jenkins pipeline.
- Used JIRA for issue tracking and Confluence for documentation.

Environment: Java 8/11, J2EE, Servlets, JSP, Java Beans, Spring, Node.js, Spring Boot, Microservices, Hibernate, RESTful services, XML, JSON, JSTL, MongoDB, AWS, Kubernetes, HTML5, CSS3, JavaScript, AJAX, Apache Tomcat, WebLogic Server, Oracle, PL/SQL, SQL, HQL, JUnit, Log4j, Continuous Integration, Jenkins, ANT, Git, JIRA, Eclipse.

EKbana Solutions, Kathmandu -Nepal
August 2019 – June 2023 | Jr. Java Developer

- Participated in the complete Software Development Life Cycle (SDLC), covering Requirement Analysis, Design, Implementation, Testing, and Maintenance.
- Developed application modules, core classes, and utility functions using core Java.
- Designed and implemented custom exception-handling mechanisms to efficiently manage application errors.
- Created Java Beans for the model layer and followed the Model-View-Controller (MVC) architectural pattern.
- Managed Terraform configurations in Git, enabling collaborative infrastructure development and version control.
- Developed client applications to interact with web services using both SOAP and REST protocols.
- Utilized Stackdriver on GCP for monitoring and logging Java application metrics, ensuring real-time performance visibility and debugging.
- Applied Dependency Injection (DI) and Inversion of Control (IOC) to retrieve bean references in the Spring framework using annotations.
- Developed applications adhering to J2EE best practices and implemented design patterns such as Singleton, Factory, Session Façade, MVC, and DAO.
- Configured VPC networks, subnets, and firewall rules on GCP to establish secure communication between Java applications and external services.
- Used Java Objects and Hibernate to build business components and facilitate object-relational mapping with databases.
- Created and optimized Oracle SQL queries and PL/SQL functions.
- Integrated Terraform scripts into Jenkins pipelines to automate infrastructure provisioning alongside Java application builds.
- Performed XSLT transformations to convert XML data into HTML format.
- Developed data-driven web applications using server-side J2EE technologies like Servlets and JSP, generating dynamic web pages with Java Server Pages (JSP).
- Mapped data representation from the MVC model to an Oracle relational database schema using Hibernate, an object-relational mapping (ORM) framework.
- Conducted thorough unit and integration testing with JUnit and Mockito to ensure software quality and reliability.
- Leveraged the Spring IOC framework to seamlessly integrate Hibernate.
- Employed Maven for dependency management, JAR packaging, and deploying applications to the WebLogic Application Server.
- Used SVN for version control and code management.
- Implemented Log4j for logging and debugging purposes.

Environment: Java/J2EE, Servlets, JSP, JSF, Struts, Hibernate, JavaBeans, XML, XSL, HTML, DHTML, JavaScript, JDBC, Log4J, Oracle, IBM Web Sphere Application Server , Windows.

EDUCATION

Masters of Science in Applied Computer Science, Northwest Missouri State University.